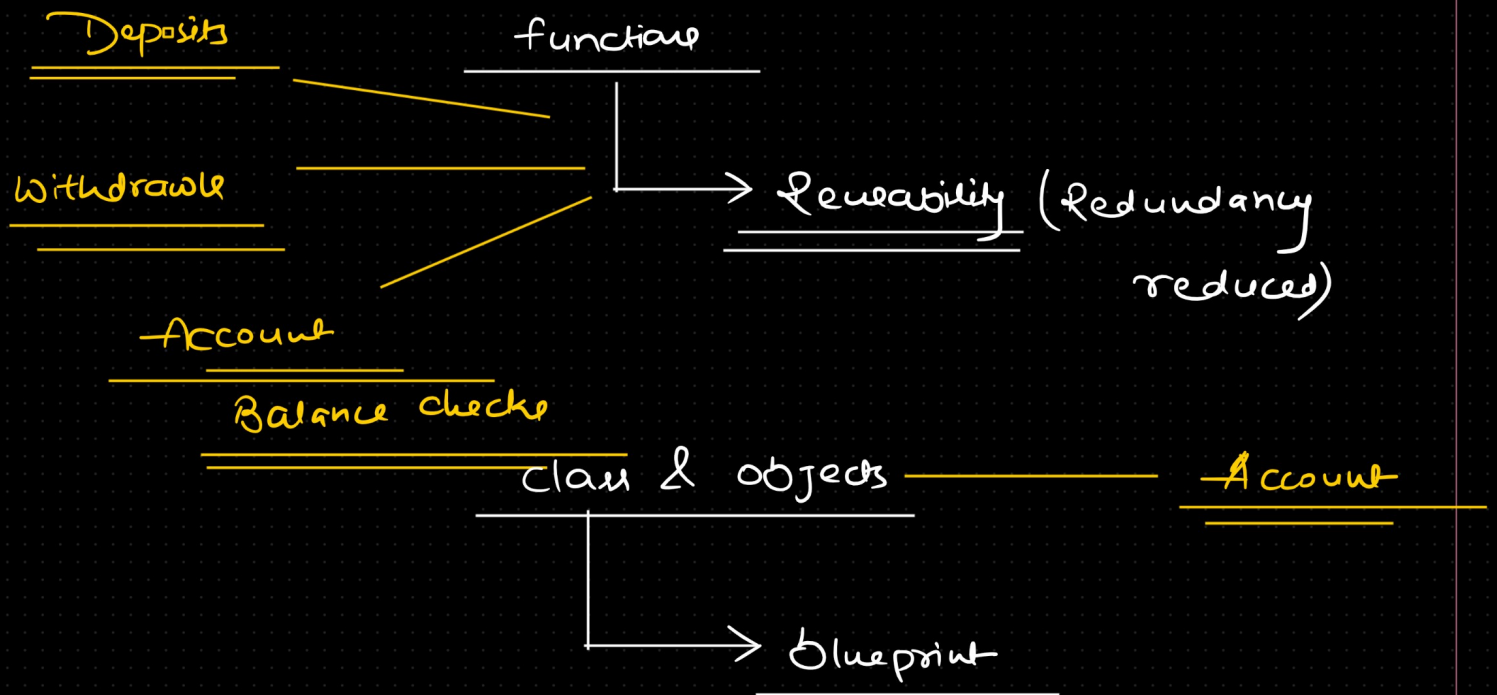


Object Oriented Programming

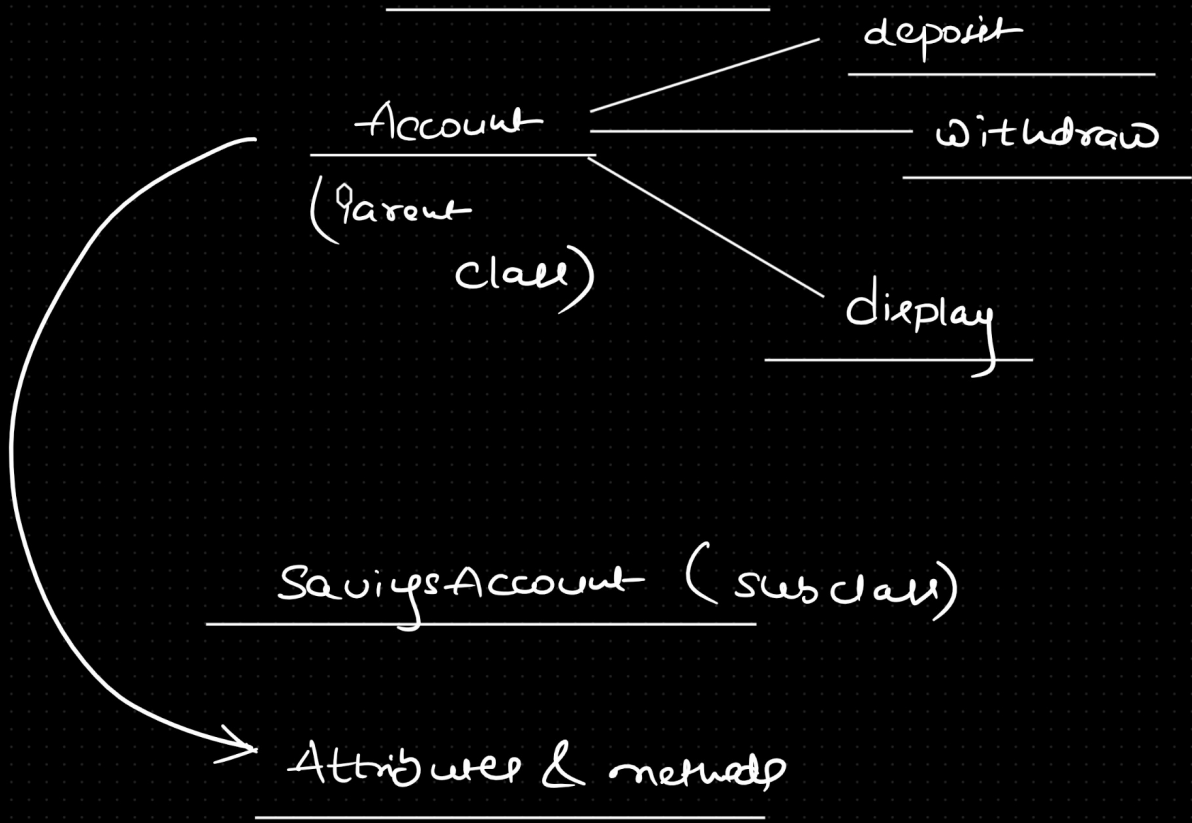


① Reusability

② Managing complexity

③ Extensibility

Inheritance ✓



Polymorphism

forms

many

Base class

class Shape:

✓ def area(self):

pass

Subclass Circle

class Circle(Shape):

def __init__(self, radius):

self.radius = radius

✓ def area(self):

return 3.14 * self.radius * self.radius

Subclass Rectangle

class Rectangle(Shape):

def __init__(self, width, height):

self.width = width

self.height = height

✓ def area(self):

return self.width * self.height

An overdraft limit in a current account is a pre-approved credit line that allows you to withdraw more money than your current account balance, up to a certain limit

Deposited 20000. New balance is 30000.

Interest added: 900.0. New balance: 30900.0

Savings account Number: SA123, Balance: 30900.0, Interest Rate: 3%

Deposited 50000. New balance is 150000.

Current Account: CA345, Balance: 150000, Overdraft Limit: 25000

Withdrew 125000. New balance: 25000

acc1

acc2

Encapsulation

class Account:

Constructor to initialize account details

```
def __init__(self, account_number, account_holder, balance):
    self.account_number = account_number
    self.account_holder = account_holder
    self.balance = balance
```

Add the money to the account

```
def deposit(self, amount):
    """Deposit money into the account."""
    if amount > 0:
        self.balance += amount
        print(f"Deposited {amount}. New balance is {self.balance}.")
    else:
        print("Deposit amount must be positive.")
```

Withdraw the money from the account

```
def withdraw(self, amount):
    if amount <= self.balance and amount > 0:
        self.balance -= amount
        print(f"{amount} withdrawn")
    else:
        print("Insufficient Funds")
```

Display the account details

```
def display(self):
    print(f"Account Number: {self.account_number}, Balance: {self.balance}")
```

Inheritance: SavingsAccount inherits from Account

class SavingsAccount(Account):

```
def __init__(self, account_number, account_holder, balance, interest_rate):
    super().__init__(account_number, account_holder, balance)
    self.interest_rate = interest_rate
```

Polymorphism: Display the account details

```
def display(self):
    print(f"Savings account Number: {self.account_number}, Balance: {self.balance}, Interest Rate: {self.interest_rate}%")
```

Added the interest in the savings account

```
def add_interest(self):
    interest = self.balance * self.interest_rate / 100
    self.balance += interest
    print(f"Interest added: {interest}. New balance: {self.balance}")
```

Inheritance: CurrentAccount inherits from Account

Overdraft Limit: An overdraft limit in a current account is a pre-approved credit

line that allows you to withdraw more money than your current account balance, up to a certain limit

class CurrentAccount(Account):

```
def __init__(self, account_number, account_holder, balance, overdraft_limit):
    super().__init__(account_number, account_holder, balance)
    self.overdraft_limit = overdraft_limit
```

Polymorphism: Overriding the withdraw method

```
def withdraw(self, amount):
    if amount <= self.balance + self.overdraft_limit:
        self.balance -= amount
        print(f"Withdrew {amount}. New balance: {self.balance}")
```

else:

```
print("Overdraft limit exceeded")
```

Polymorphism: Overriding the display method

```
def display(self):
```

```
    print(f"Current Account: {self.account_number}, Balance: {self.balance}, Overdraft Limit: {self.overdraft_limit}")
```

```
if __name__ == '__main__':
```

```
    acc1 = SavingsAccount("SA123", "Priya Bhatia", 10000, 3)
```

```
    acc2 = CurrentAccount("CA345", "Shivani Gupta", 100000, 25000)
```

```
    acc1.deposit(20000)
```

```
    acc1.add_interest()
```

```
    acc1.display()
```

```
    acc2.deposit(50000)
```

```
    acc2.display()
```

```
    acc2.withdraw(125000)
```

Abstraction

Abstraction in Python is a core principle of object-oriented programming (OOP) that simplifies complex systems by focusing on essential characteristics while hiding unnecessary details.

Deposited 20000. New balance is 30000.

Interest added: 900.0. New balance: 30900.0

Savings account Number: SA123, Balance: 30900.0, Interest Rate: 3%

Deposited 50000. New balance is 150000.

Current Account: CA345, Balance: 150000, Overdraft Limit: 25000

Withdrew 125000. New balance: 25000